Deflection Analysis - Backcalculations

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Benefits of Deflection Data Analysis

- The purpose of this work is to analyse the structural capacity of the pavement network
- Classify the highway network according to its structural condition.
- Assess pavement layers moduli (for overlay design and assessing deterioration)-Project Level
- Determine the remaining service life of the pavement network – Network Level
- Identify areas with weak subgrade or drainage problems
- Assess the optimum rehabilitation and maintenance strategies to keep the network within acceptable performance level

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Pavement Condition Data

- **Surface defects**
  Cracking, potholes, and roughness
  These data are good indicators of the functional performance (riding quality) and they are related somehow to the structural capacity of the road BUT not always.
- **Safety related data**
  The micro-texture and macro-texture determine pavement skid resistance.
- **Deflection Data**
  Deflection is an excellent indicator of the pavement structural capacity because it represents pavement response to load.

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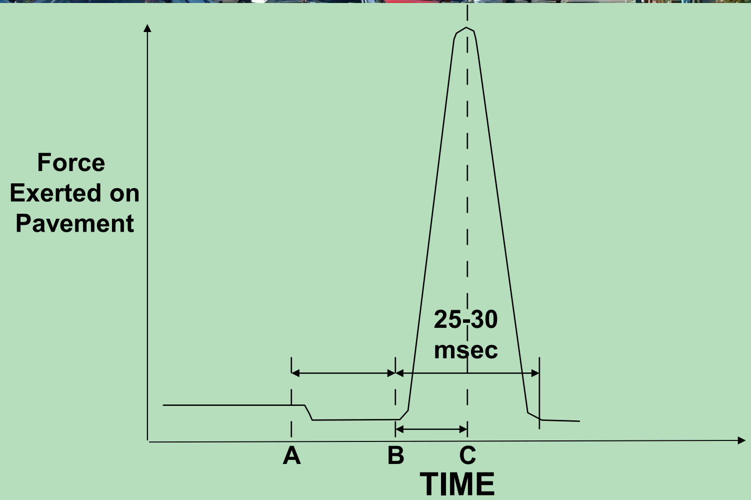
Impulse load (FWD)



(Time from A to B is Variable, Depending on Drop Height)

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Measuring Deflection by Falling Weight Deflectometer

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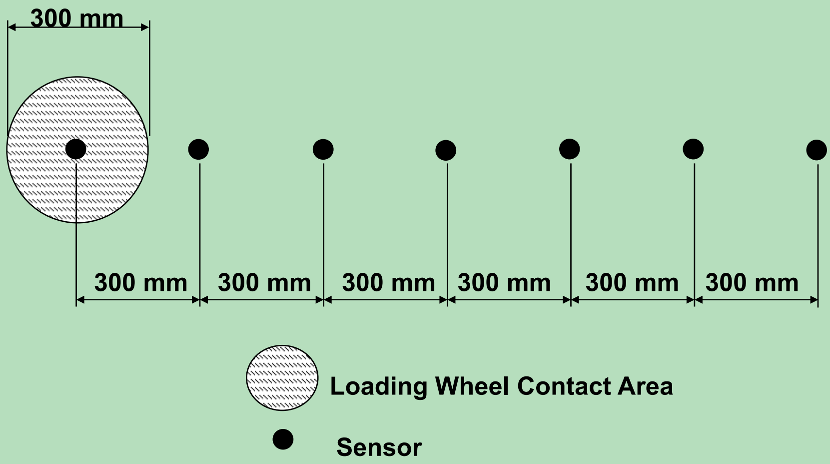

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FWD basic concept

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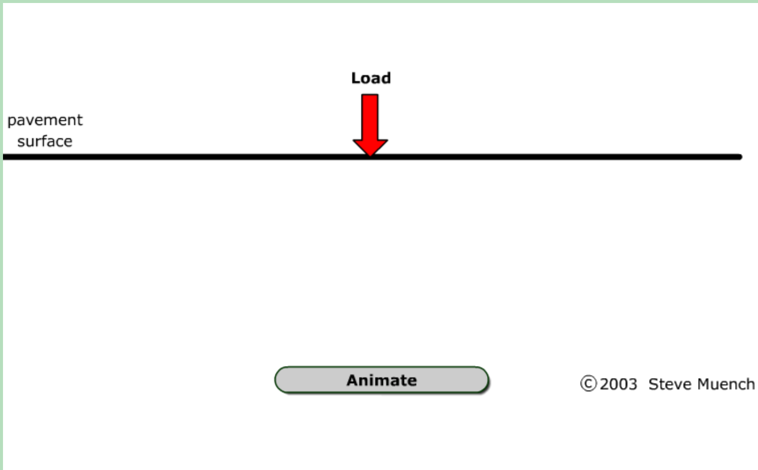

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Deflectograph

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How can we make the best out of these numbers?

152_00_ 8.45_PD.	TRUCK	ID:	Pitt&sherry														
PRP1	-	Int	150A.	LANE:	South	Bound.											
PER	BOWL																
DATA	FOR	LEFT	WHEEL.														
IS	BOWL	NO.	ODOM	DISTANCE	AND	22	DEFLECTIONS.										
DISTANCE	D0	D50	D100	D150	D200	D250	D300	D350	D400	D450	D500	D550	D600	D650	D700	D750	D800
0.01472	0.25	0.24	0.22	0.18	0.14	0.11	0.09	0.07	0.05	0.04	0.03	0.02	0.01	0	0	0	0
0.01858	0.18	0.17	0.14	0.12	0.1	0.08	0.07	0.05	0.04	0.03	0.02	0.01	0	0	0	0	0
0.02225	0.22	0.22	0.2	0.18	0.15	0.13	0.11	0.09	0.08	0.07	0.06	0.05	0.04	0.02	0.01	0	0
0.02601	0.26	0.26	0.24	0.21	0.18	0.14	0.12	0.1	0.08	0.07	0.05	0.03	0.02	0.01	0.01	0	0
0.02968	0.24	0.23	0.19	0.15	0.11	0.08	0.06	0.04	0.03	0.02	0.01	0	0	0	0	0	0
0.03344	0.07	0.06	0.05	0.04	0.03	0.02	0.02	0.02	0.01	0.01	0	0	0	0	0	0	0
0.03716	0.18	0.18	0.16	0.14	0.11	0.08	0.06	0.05	0.04	0.04	0.03	0.03	0.02	0.01	0	0	0
0.04088	0.13	0.13	0.12	0.09	0.07	0.04	0.03	0.02	0.02	0.01	0.01	0	0	0	0	0	0
0.04454	0.16	0.16	0.14	0.13	0.11	0.09	0.07	0.06	0.04	0.03	0.02	0.02	0.01	0.01	0	0	0
0.04826	0.23	0.23	0.21	0.18	0.16	0.13	0.11	0.1	0.08	0.07	0.05	0.04	0.03	0.02	0.01	0	0
0.05197	0.26	0.25	0.23	0.2	0.17	0.14	0.11	0.08	0.07	0.07	0.06	0.05	0.05	0.04	0.03	0.02	0
0.05569	0.19	0.18	0.16	0.14	0.11	0.08	0.07	0.05	0.04	0.02	0.01	0.01	0.01	0	0	0	0
0.0594	0.2	0.19	0.16	0.14	0.1	0.08	0.07	0.05	0.04	0.03	0.02	0.01	0.01	0	0	0	0
0.06312	0.16	0.16	0.14	0.12	0.09	0.07	0.06	0.05	0.05	0.04	0.03	0.03	0.02	0.02	0.01	0	0
0.06684	0.18	0.16	0.13	0.1	0.07	0.05	0.03	0.02	0.02	0.01	0.01	0	0	0	0	0	0
0.0706	0.19	0.18	0.16	0.12	0.09	0.06	0.05	0.04	0.03	0.02	0.02	0.01	0.01	0	0	0	0
0.07432	0.27	0.26	0.24	0.21	0.18	0.15	0.12	0.1	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0
0.07808	0.17	0.16	0.13	0.1	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.01	0	0	0
0.0817	0.14	0.13	0.11	0.1	0.08	0.06	0.05	0.05	0.04	0.04	0.03	0.03	0.02	0.02	0.01	0	0
0.08547	0.12	0.11	0.09	0.07	0.06	0.05	0.04	0.03	0.02	0.02	0.01	0.01	0	0	0	0	0

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Data Analysis

- How can we aggregate these data into reliable index/s?
- Reliable index must reflect the actual structural condition in the field.
- For network analysis, indices must be easy to calculate in order to practically generate them for the entire network.
- Backcalculation method is not practical at the network level.

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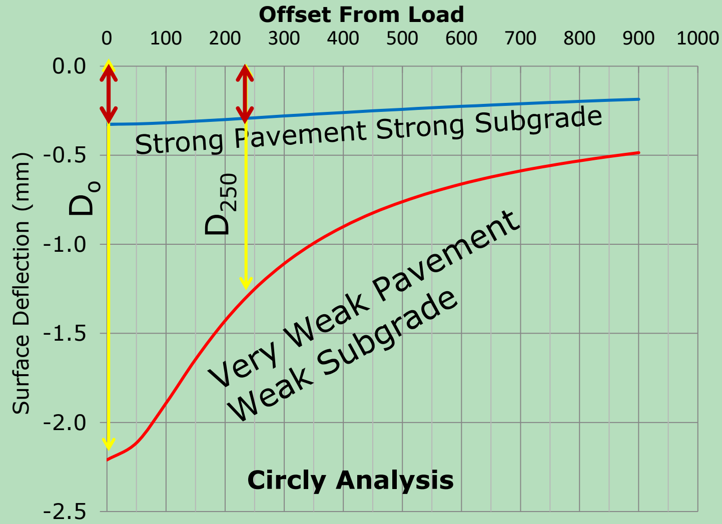
Deflection Basin for Weak and Strong Pavements



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Deflection Basin Shape and Pavement Structural Capacity

- Weak pavements structures over weak subgrade tend to have large localise deflection under the load but the deflection diminishes quickly.
- Strong pavements structures will show small deflection under the load and the deflection is nearly constant for large distance from the load. This is because of the high shear strength and the spread of the load over a wide area.

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Deflection Basin Parameters

- Maximum Central Deflection (D_0) this parameter is related to the structural capacity of the entire pavement and it is good measure of the subgrade bearing capacity.
- Surface Curvature (D_0-D_{200}): this is a measure of the tensile strains at the bottom of the asphalt surface and therefore good indicator of asphalt fatigue.
- Deflection Ratio (D_{250}/D_0): This parameter is good indicator of pavement strength and is used by TMR

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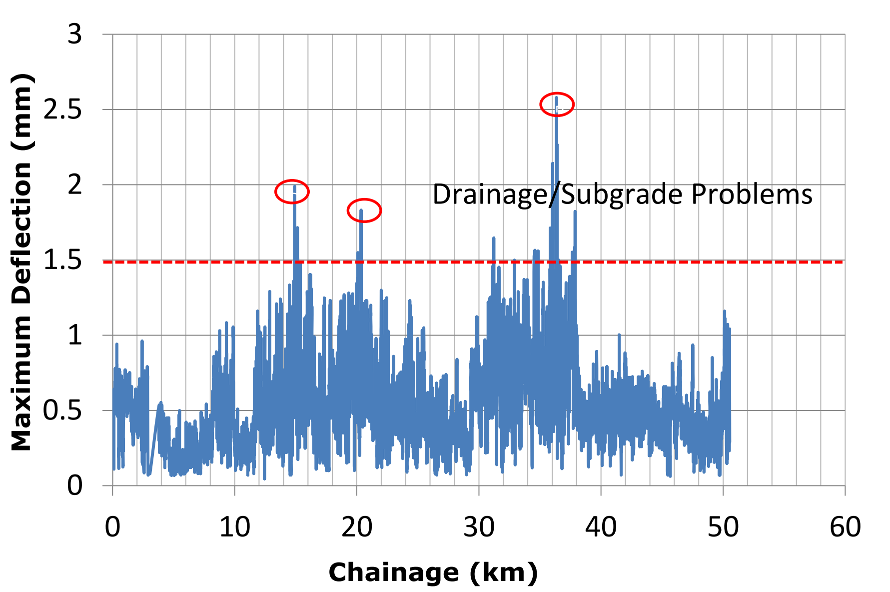
Diagnose Subgrade/Drainage Problems



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Deflection Ratio as Recommended by TMR

Deflection Ratio	Pavement condition
$D_r \geq 0.8$	indicates a bound pavement
$0.6 \leq D_r \leq 0.7$	indicates good quality unbound pavement with thin asphalt seal
$D_r < 0.6$	indicates a possible weak unbound pavement with thin asphalts.

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- Base Layer Index (BLI)= D_0-D_{300}
- Middle Layer Index (MLI)= D_{300}-D_{600}
- Lower Layer Index (LLI)= D_{600}-D_{900}

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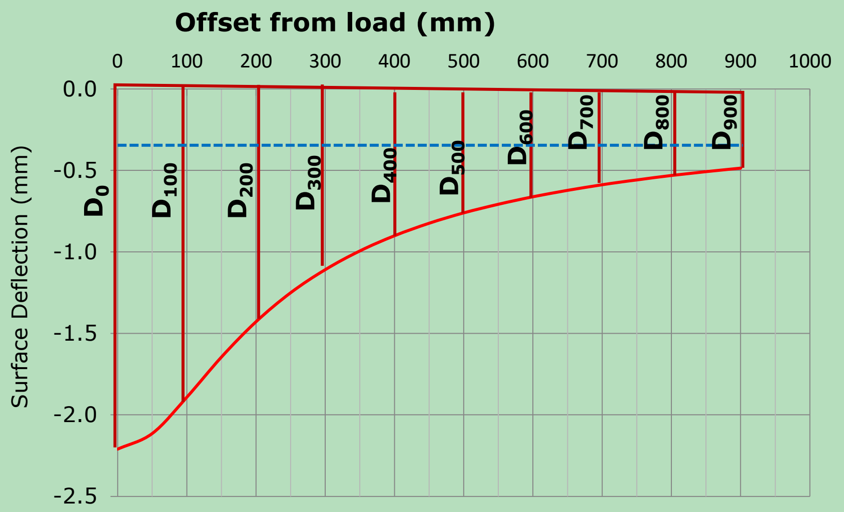
Normalised Area Parameter



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Normalised Area Parameter

- Normalised area parameter is the total area of the deflection basin normalised by the central deflection. This parameter is a good indicator of the structural integrity of the pavement structure.
- Strong pavements structures have large normalised area and vice versa.

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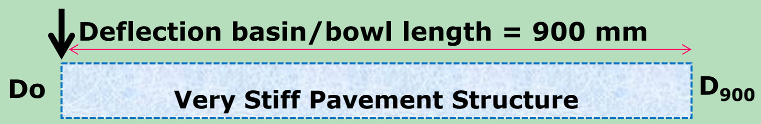
Normalised Area Parameter

$$A = \frac{50}{D_0} * \left\{ \left(\frac{D_0 + D_{50}}{2} \right) + \left(\frac{D_{50} + D_{100}}{2} \right) + \left(\frac{D_{100} + D_{150}}{2} \right) + \ldots\ldots\ldots \left(\frac{D_{850} + D_{900}}{2} \right) \right\}$$

$$A = \frac{50}{D_0} \left\{ \left(\frac{D_0 + D_{900}}{2} \right) + \sum_{i=50}^{850} D_i \right\}$$

For Very strong pavement: $D_0 = D_{50} = \ldots\ldots D_{900}$
A=900 mm^2/mm for very stiff pavement structure



$A = D_0 * 900 / D_0 = 900$ mm^2/mm

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- The area value tends to provide a fairly good indication of the relative stiffness of the pavement section, particularly the bound layer, because it is largely insensitive to subgrade stiffness. Combining the area value in a plot with E_{SG} and D_o provides a good picture of the relative stiffness of both the surfacing and subgrade that interact to produce the measured deflections.

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FWD		**Generalized Conclusions**
Area	**Do**	
Low	Low	**Weak** pavement structure but **strong** subgrade
Low	High	**Weak** pavement structure but **Weak** subgrade
High	Low	Strong Pavement structure and Strong Subgrade
High	High	Strong Pavement structure but weak subgrade

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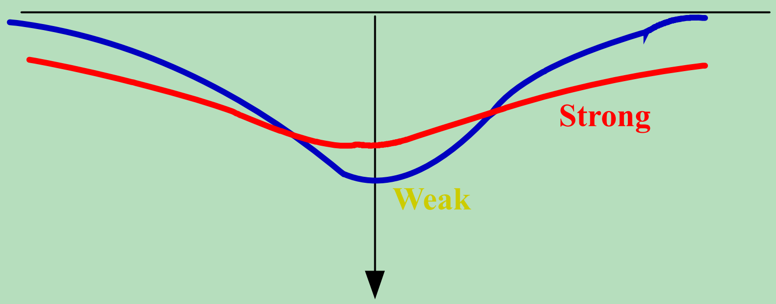

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Area Ratio Parameter

- It is defined as the ratio between the normalised area of any pavement section to the normalised area of the stiffest pavement structure.

$$A_r = \frac{50}{900 * D_0}\left\{\left(\frac{D_0 + D_{900}}{2}\right) + \sum_{i=50}^{850} D_i\right\}$$

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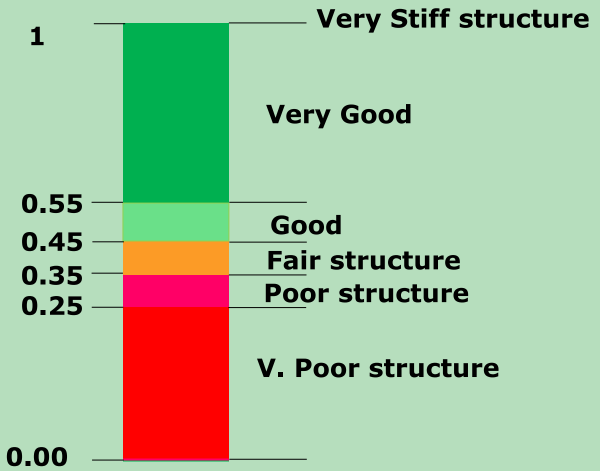
Area Ratio Scale



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Classification of the Network based on Area Ratio

Area Ratio (A_r)	Road Condition	Observed Distress
≤0.25	Very Poor	Severe Fatigue cracks with high intensity, deep ruts
$0.25<A_r\leq0.35$	Poor	Moderate to high fatigue cracks, deep ruts
$0.35<A_r\leq0.45$	Fair	moderate fatigue cracks and moderate ruts
$0.45<A_r\leq0.55$	Good	Some minor cracks and/or shallow ruts
$0.55<A_r\leq0.65$	Very Good	Some minor cracks and/or shallow ruts
0.65<Ar	Excellent	Very minor

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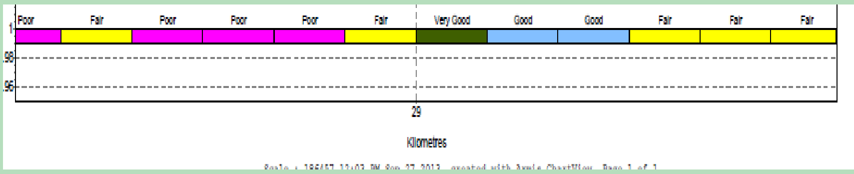
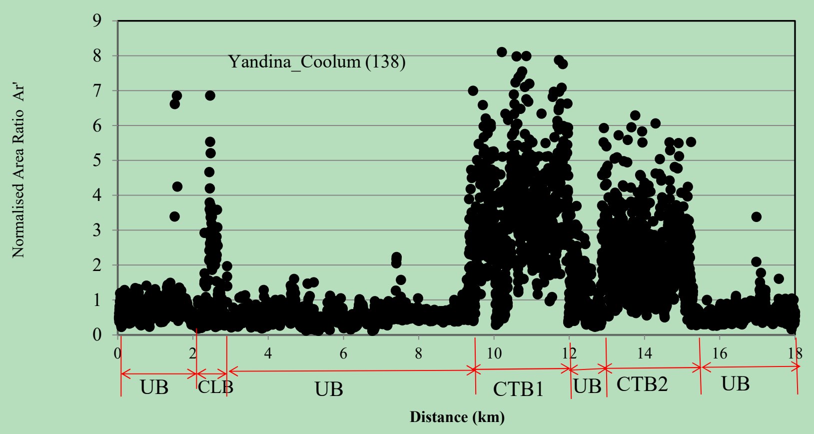
Colour Coded Network



Normalised Area Ratio

$$Normalised \quad A_r = \frac{50}{900 * D_0^2}\left\{\left(\frac{D_0 + D_{900}}{2}\right) + \sum_{i=50}^{850} D_i\right\}$$

Normalised Area Ratio



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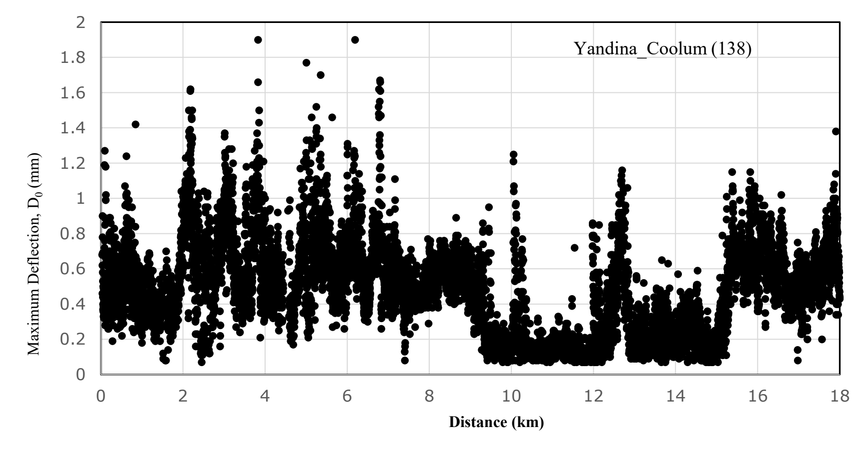
D_0 Versus Chainage



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Backcalculation

- **Forward-Calculation:** in this process, the pavement moduli, thicknesses, Poisson ratio, and loads are known and the pavement responses are required.
- **Backcalculation:** In this process the pavement response (deflection basin), thicknesses, loads are known and the layers moduli are required.

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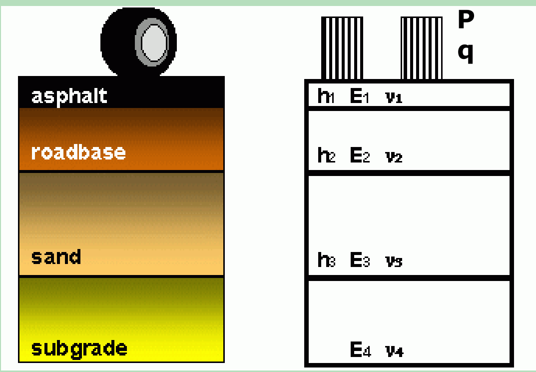
Forward Calculation



Circly analysis or any other Multilayer Elastic Analysis (MEA) software
Everstress
WinJulea
Bisar

Input: Load and pavement properties
Pavement Response: Stresses, Strains and Deflections

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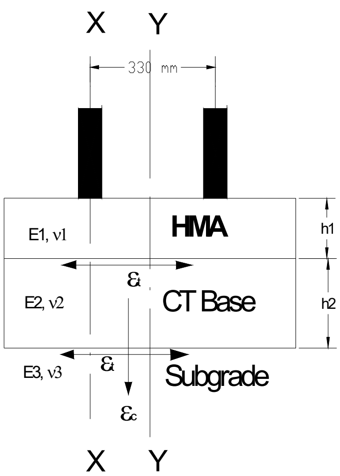
In forward analysis:
Determine pavement response, stresses, strain, deflection



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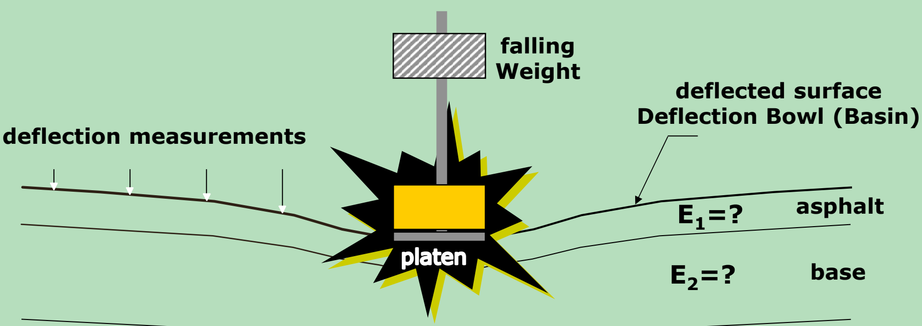
Backcalculation based on Non destructive FWD Test



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Falling Weight Deflectometer (FWD)

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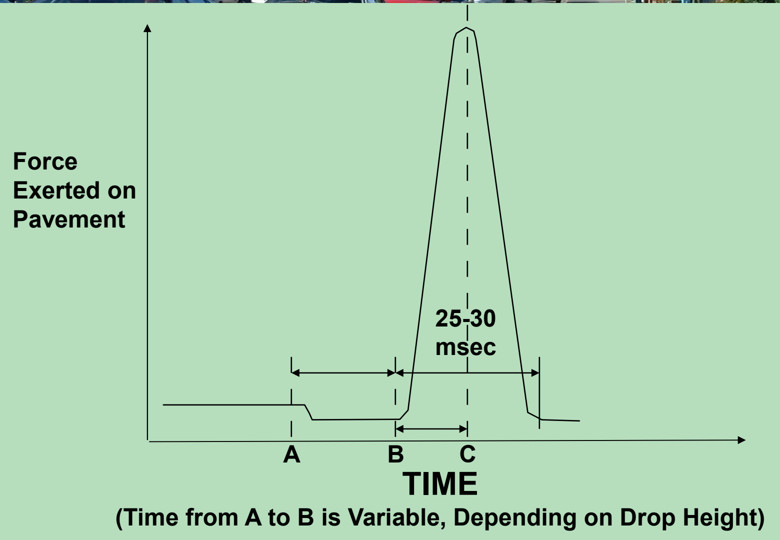
Impulse load (FWD)



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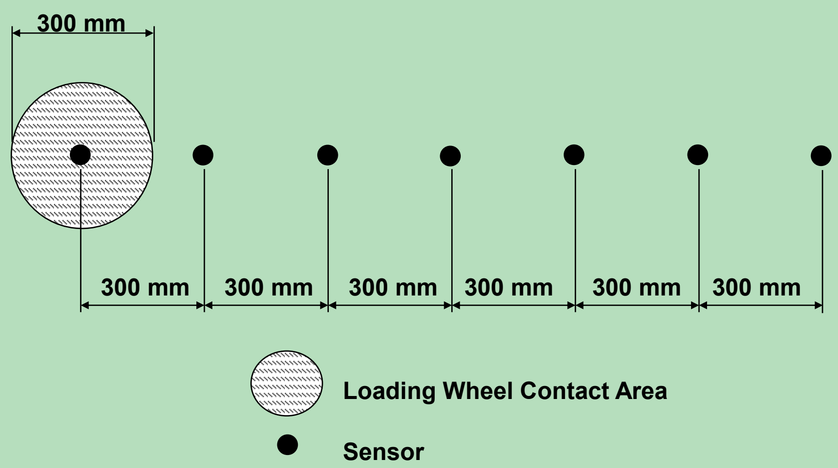
Sample FWD sensor spacing



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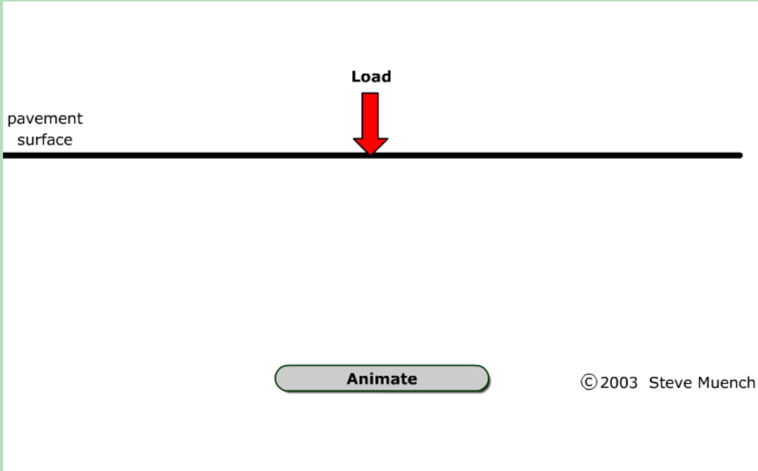
FWD basic concept



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- **Backcalculation**
- Involves estimating pavement layer moduli from measured surface deflections, and , usually known layer thicknesses.
- The moduli can be used in:
- Fundamental engineering analyses
- As an indication of material condition.

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- **Most backcalculation procedures are based on the assumption that deflection measured at some relatively large distance from the load primarily reflect subgrade response**. This allows use of the outer sensor deflections for estimating subgrade modulus as a starting point, and the solution for all layers is typically derived iteratively from there.
- Deflection under the centre of the load which represents the total deflection of the pavement.

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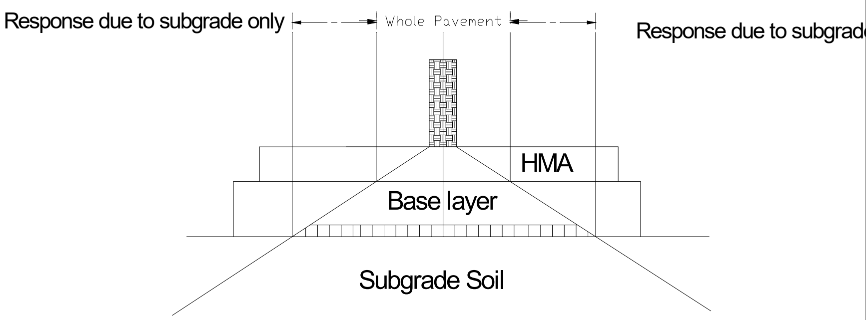

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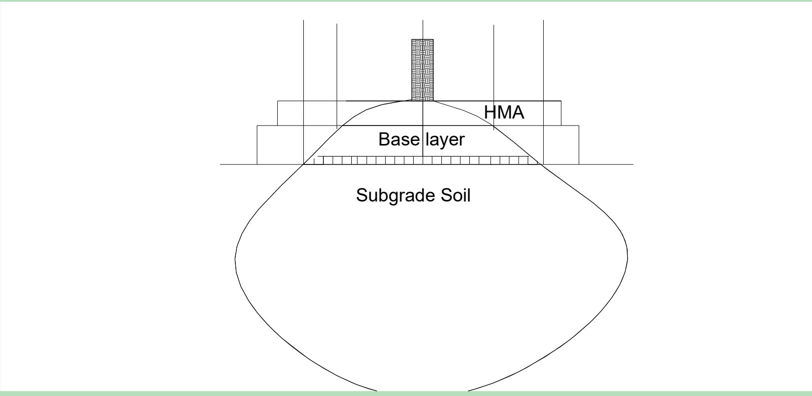

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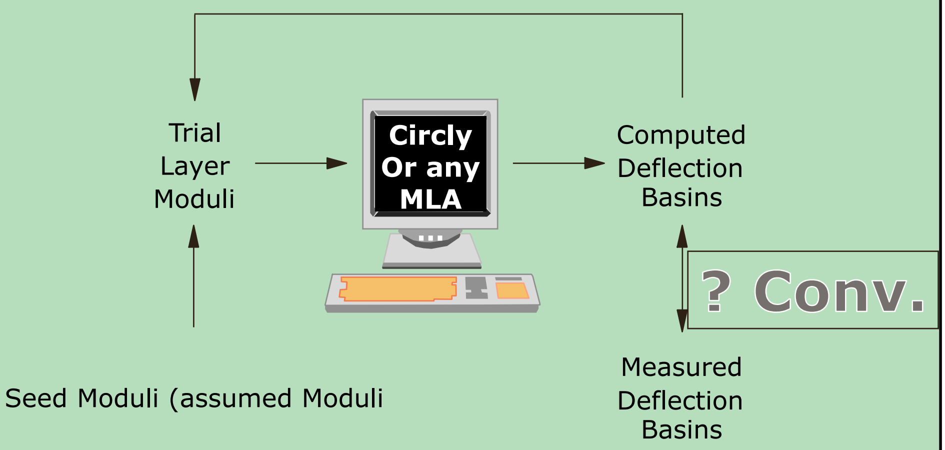
Iteration Approach



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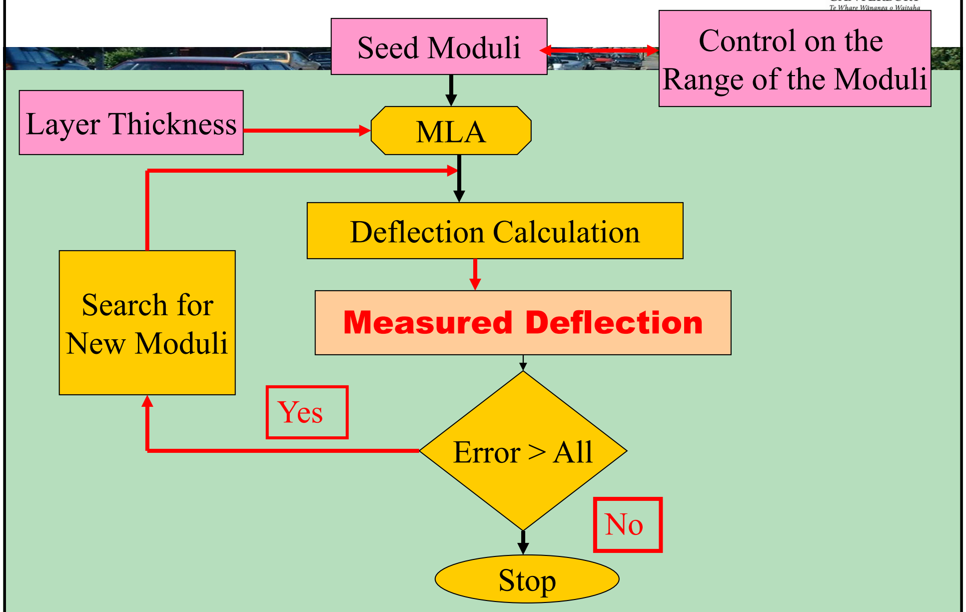
Backcalculation



Measures of deflection basin convergence

Average of absolute relative differences (ABS)

$$ABS = \frac{1}{n} * \sum_{i=1}^{n} \left(\left| \frac{d_{ci} - d_{mi}}{d_{mi}} \right| \right) * 100$$

Where,
ABS = average of absolute relative differences between the calculated and measured deflection basin, expressed as a percentage.
d_{ci}= Calculated pavement surface deflection at sensor i,
d_{mi} = measured pavement surface deflection at sensor I, and
n= number of deflection sensors used in the backcalculation process.

Root Mean Square (RMS)

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$$RMS = \left(\sqrt{\frac{1}{n} * \sum_{i=1}^{n} \left(\frac{d_{ci} - d_{mi}}{d_{mi}} \right)^2} \right) * 100$$

where,
RMS = Root mean square error
d_{ci}= Calculated pavement surface deflection at sensor i,
d_{mi} = measured pavement surface deflection at sensor I, and
n= number of deflection sensors used in the backcalculation process.

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Sum of Absolute Values of the Relative Differences (ARS)

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$$ARS\% = \sum_{i=1}^{n} \left(\left| \frac{d_{ci} - d_{mi}}{d_{mi}} \right| \right) * 100$$

where,
ARS = Sum of the absolute values of relative differences between the calculated and measured deflection basin, expressed as a percentage
d_{ci}= Calculated pavement surface deflection at sensor i,
d_{mi} = measured pavement surface deflection at sensor I, and
n= number of deflection sensors used in the backcalculation process

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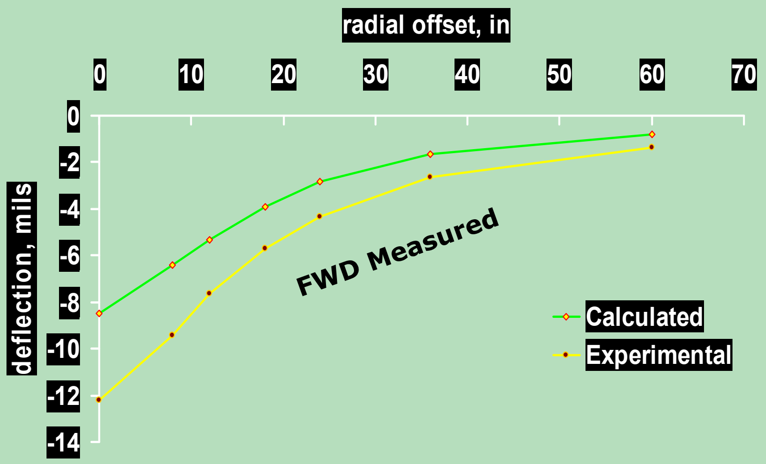
Iterations to Match Measured and calculated deflection Basin



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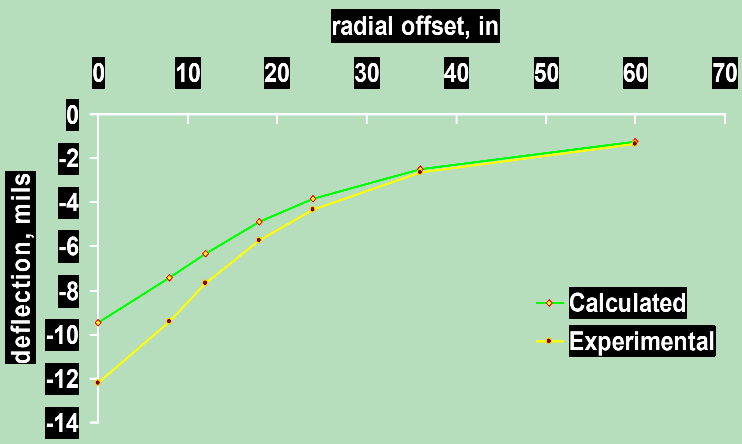
Iterations to Match Measured and calculated deflection Basin



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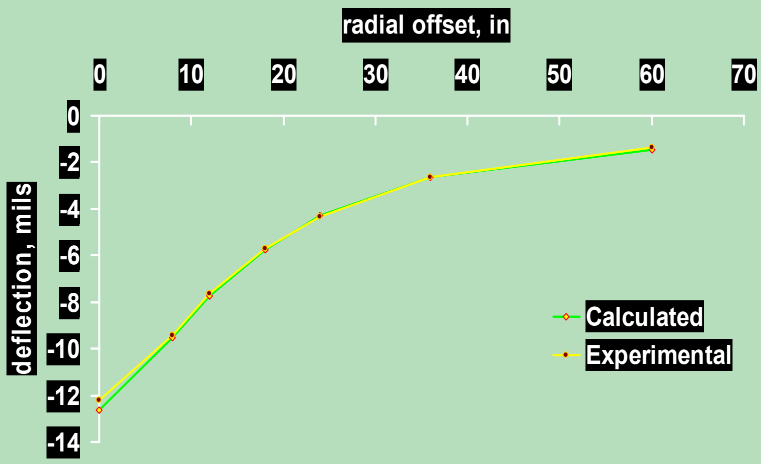
Iterations to Match Measured and calculated deflection Basin



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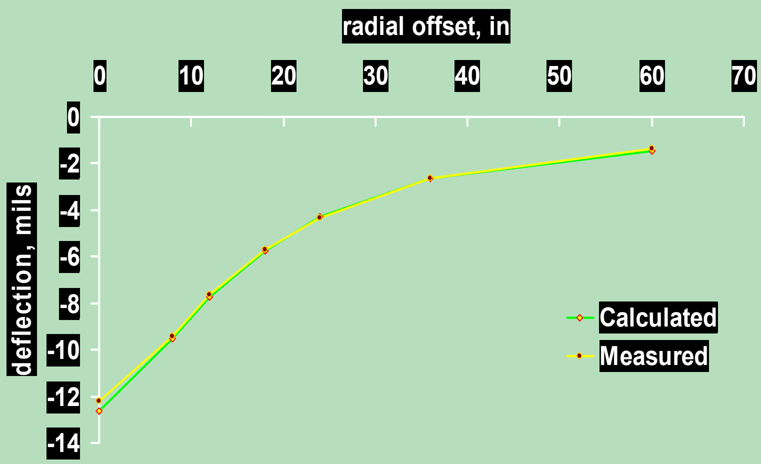
Iterations to Match Measured and calculated deflection Basin



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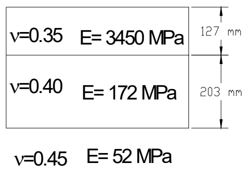
Example (Manual Backcalculation)

- Forward Analysis



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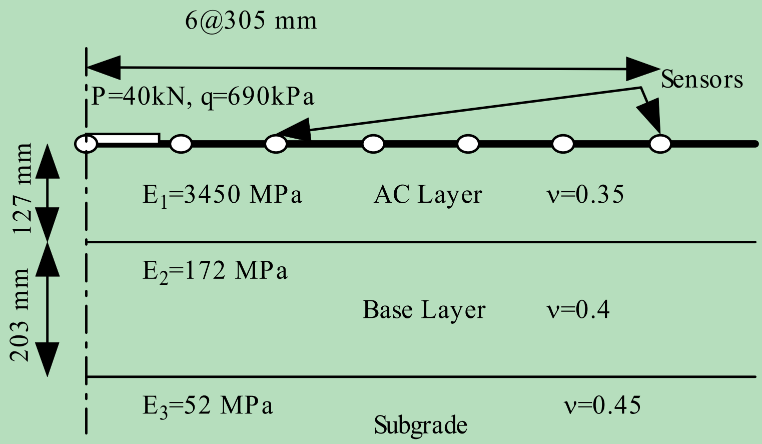

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Deflection results from the forward analysis

Sensor	Do	D1	D2	D3	D4	D5	D6
Offset (cm)	0	30.5	61.0	91.5	122	152.5	183
Deflection	689	515	346	240	176	136	110

Now assume we have the deflection results and we don't know the pavement layers moduli.

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Backcalculation

- Step #1
- Assume seed moduli for the different layers.
- For subgrade use the following equation and the deflection results at the farthest sensor from the load.

$$M_r = \frac{P*(1-\nu^2)}{\pi * D_r * r}$$

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- M_r= Backcalculated subgrade resilient modulus
- P = Applied load
- D_r= Pavement surface deflection at a distance r from the centre of the load plate
- r = Distance from centre of load plate to D_r.

$$E_{sub} = \frac{40 * (1 - 0.45^2)}{\pi * 0.11 * 10^{-3} * 1.830} = 50.4 \quad MPa$$

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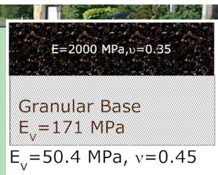
Iteration #1

- Assume E_{base} =170 MPa
- E_{AC}= 2000 MPa



- Now Run forward analysis and calculate pavement deflections.

Sensor	Do	D1	D2	D3	D4	D5	D6
Measured Def	**689**	**515**	**346**	**240**	**176**	**136**	**110**
Calculated Def	**793**	**557**	**359**	**245**	**179**	**139**	**113**
Difference	-104	-42	-13	-5	-3	-3	-3

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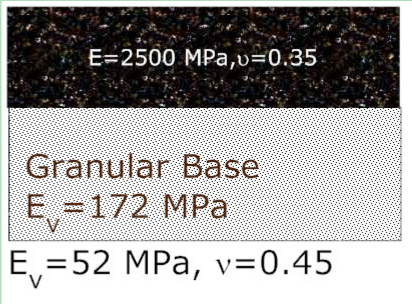
- Trail #2
- Increase the modulus of asphalt layer because the deflection under the load is higher than the targeted deflection.
- Assume E_{AC} = 2500 MPa
- Assume E_b=172 MPa
- E_{sub}= 52 MPa



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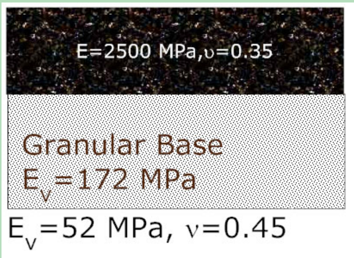
Iteration #2

Sensor	Do	D1	D2	D3	D4	D5	D6
Measured Def	**689**	**515**	**346**	**240**	**176**	**136**	**110**
Calculated Def	**740**	**532**	**348**	**239**	**174**	**135**	**110**
Difference	-51	-17	-2	1.0	2.0	1.0	0



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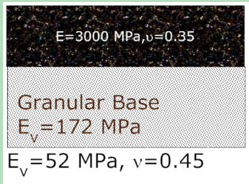
Iteration #3

- Trial # 3
- Again the problem still in the AC modulus. Increase E_{AC} = 3000 MPa
- Assume E_b=172 MPa
- E_{sub}= 52 MPa



Sensor	Do	D1	D2	D3	D4	D5	D6
Measured Def	**689**	**515**	**346**	**240**	**176**	**136**	**110**
Calculated Def	**711**	**523**	**347**	**240**	**175**	**136**	**110**
Difference	-22	-8	-1	0	1	0	0

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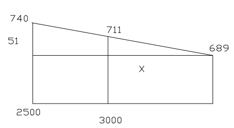
- Trial #4
- Using Linear Interpolation
- EAC = 2500 ----- Do= 740
- EAC = ------ Do =689
- EAC= 3000 Do = 711



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$$\frac{x}{500 + x} = \frac{22}{51}$$

E_{AC}=3380 MPa

Trial #4

- Increase E_{AC} = 3380 MPa
- Assume E_b=172 MPa
- E_{sub}= 52 MPa

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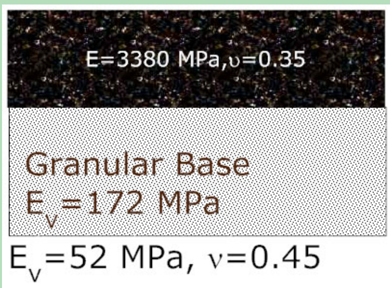
Iteration #3

Sensor	Do	D1	D2	D3	D4	D5	D6
Measured Def	689	515	346	240	176	136	110
Calculated Def	692	516	346	240	176	136	110
Difference	-3	-1	0	0	0	0	0



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- C:\Program Files\pavement\EVERSERS\EverCalc
- BAKFAA from Federal Aviation Administration website

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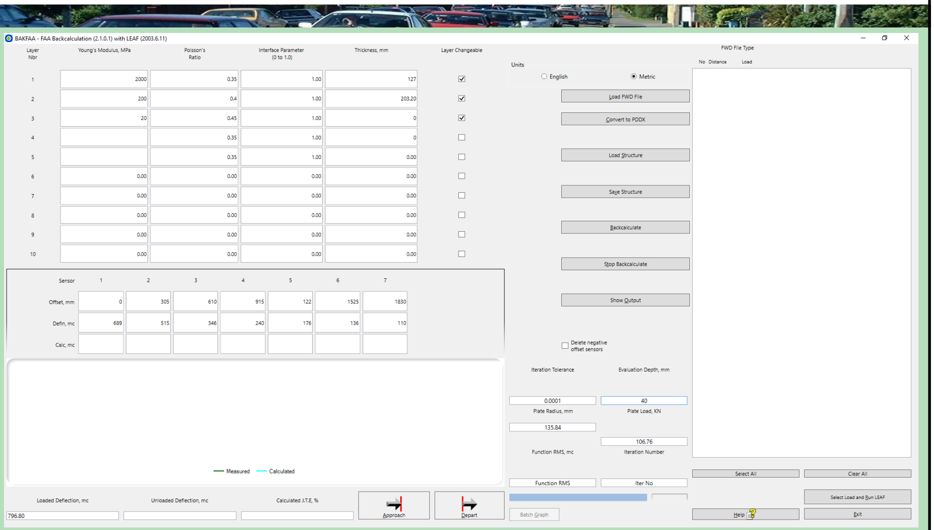
BAKFAA Software



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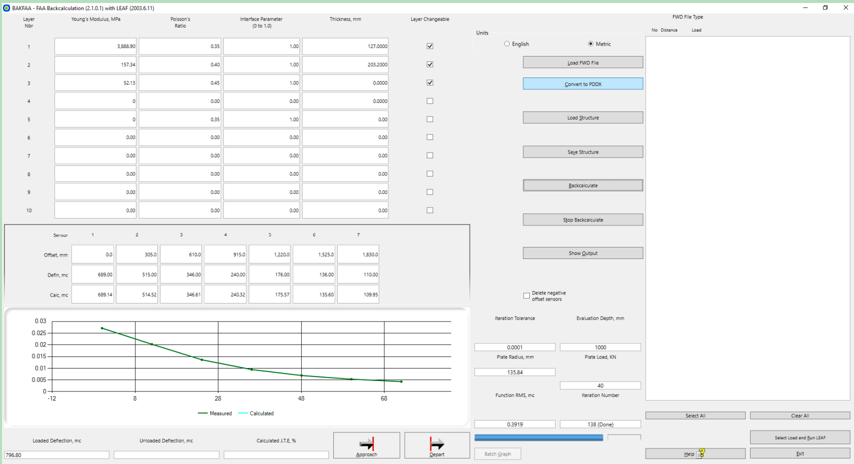
BAKFAA - FAA Backcalculation (2.1.0.1) with LEAF (2003.6.11)
Layer Nbr
Young's Modulus, MPa
Poisson's Ratio
Interface Parameter (0 to 1.0)
Thickness, mm
Layer Changeable
1 3,888.90 0.35 1.00 127.0000
2 157.34 0.40 1.00 203.2000
3 52.13 0.45 1.00 0.0000
4 0 0.00 0.00 0.0000
5 0 0.35 1.00 0.00
6 0.00 0.00 0.00 0.00
7 0.00 0.00 0.00 0.00
8 0.00 0.00 0.00 0.00
9 0.00 0.00 0.00 0.00
10 0.00 0.00 0.00 0.00
Sensor 1 2 3 4 5 6 7
Offset, mm 0.0 305.0 610.0 915.0 1,220.0 1,525.0 1,830.0
Defln, mc 689.00 515.00 346.00 240.00 176.00 136.00 110.00
Calc, mc 689.14 514.52 346.61 240.32 175.57 135.60 109.95
0.03
0.025
0.02
0.015
0.01
0.005
0
-12
8
28
48
68
Measured
Calculated
Loaded Deflection, mc
796.80
Unloaded Deflection, mc
Calculated J.T.E, %
Approach
Depart
Units
English
Metric
Load FWD File
Convert to PDDX
Load Structure
Save Structure
Backcalculate
Stop Backcalculate
Show Output
Delete negative offset sensors
Iteration Tolerance
0.0001
Evaluation Depth, mm
1000
Plate Radius, mm
135.84
Plate Load, KN
40
Function RMS, mc
0.3919
Iteration Number
138 (Done)
Batch Graph
FWD File Type
No Distance Load
Select All
Clear All
Select Load and Run LEAF
Help
Exit

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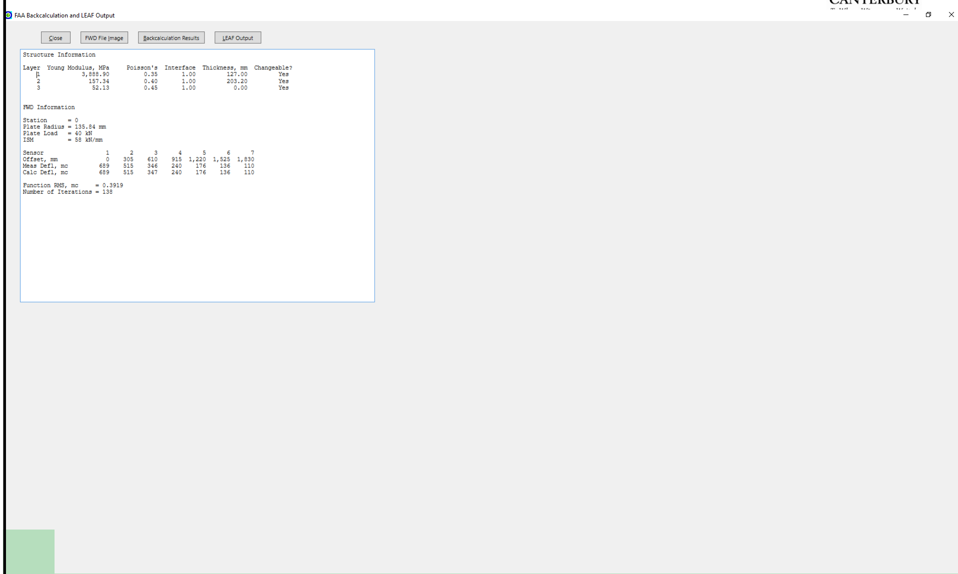
FAA Backcalculation and LEAF Output
Close
FWD File Image
Backcalculation Results
LEAF Output
Structure Information
Layer Young Modulus, MPa Poisson's Interface Thickness, mm Changeable?
1 3,888.90 0.35 1.00 127.00 Yes
2 157.34 0.40 1.00 203.20 Yes
3 52.13 0.45 1.00 0.00 Yes
FWD Information
Station = 0
Plate Radius = 135.84 mm
Plate Load = 40 kN
ISM = 58 kN/mm
Sensor 1 2 3 4 5 6 7
Offset, mm 0 305 610 915 1,220 1,525 1,830
Meas Defl, mc 689 515 346 240 176 136 110
Calc Defl, mc 689 515 347 240 176 136 110
Function RMS, mc = 0.3919
Number of Iterations = 138

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Output BAKFAA

- Layer Young Modulus, MPa Poisson's Interface Thickness, mm Changeable?
- 1 3,888.90 0.35 1.00 127.00 Yes
- 2 157.34 0.40 1.00 203.20 Yes
- 3 52.13 0.45 1.00 0.00 Yes

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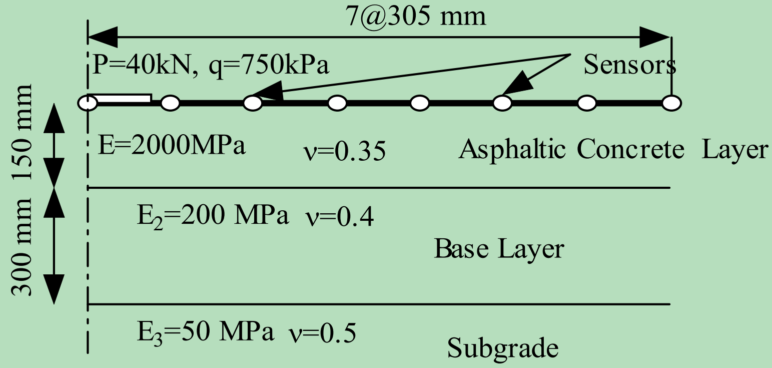

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offset	0	30.5	61	91.5	122	152.5	183	213.5
Def	653.9	464.8	324.3	236.7	179.4	140.8	114.2	95.41

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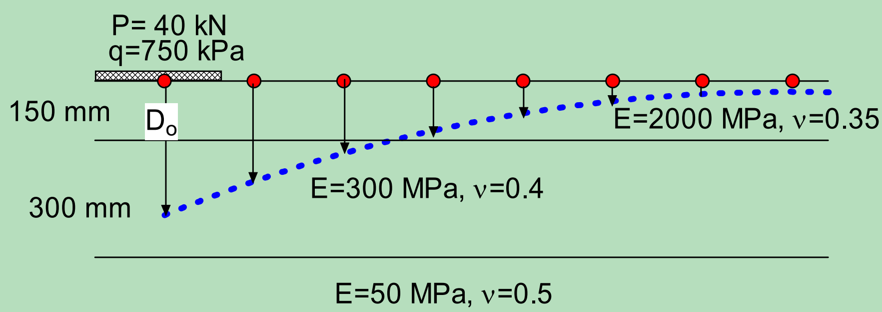
Example



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Example

Offset	0	30.5	61	91.5	122	152.5	183	213.5
Def	587.5	422.2	307.8	232.8	180.5	143.3	116.7	97.42

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Sample text slide

- This is a sample text slide
  - With multiple levels of text

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